

Erik Jonsson School of Engineering and Computer Science

Department of Computer Science

Computer Science (BS)

The Computer Science Department offers the [BS degree in Computer Science](#) and the [BS degree in Software Engineering](#). Both are based on a solid foundation of mathematics, including calculus, and discrete mathematics. These programs of study are designed to offer students opportunities to prepare for an industrial, business, or governmental career in a rapidly changing profession and to prepare for graduate study in a field in which further education is strongly recommended. The two programs have the same basis in core computer science, including the analysis of algorithms and data structures, modern programming methodologies, and the study of operating systems. The Computer Science program continues with courses in advanced data structures, programming languages, telecommunications networks, and automata theory, while the Software Engineering program includes courses in requirements engineering, software validation and testing, and software architecture, culminating in a challenging project course in which students must demonstrate use of software engineering techniques. Both programs offer a rich choice of elective studies, including courses in artificial intelligence, computer graphics, and compiler design.

The school offers a "fast track" BS/MS option; see [Fast Track Baccalaureate/Master's Degree Program](#).

The Computer Science Department and the Department of Mathematical Sciences in the School of Natural Sciences and Mathematics jointly offer a [BS degree in Data Science](#).

Mission of the Department of Computer Science

The mission of the Department of Computer Science is to prepare undergraduate and graduate students for productive careers in industry, academia, and government by providing an outstanding environment for teaching, learning, and research in the theory and applications of computing. The Department places a high priority on establishing and maintaining innovative research programs to enhance its education quality and make it an important regional, national, and international resource center for discovering, integrating, and applying new knowledge and technologies.

Goals for the Computer Science Program

The undergraduate Computer Science program is committed to providing students with a high-quality education and preparing them for long and successful careers in industry and government.

Our graduates, while eminently ready for immediate employment, will also be fully ready for focused training as required for specific positions in Computer Science and closely related areas. Graduates interested in highly technical careers, research, and/or academia will be fully prepared to further their education in graduate school.

Program Educational Objectives for Computer Science

Within a few years after graduation, graduates of the Computer Science program should:

- Pursue lasting and successful careers in computer science or related fields including as industry professionals, educators, researchers, government professionals and/or entrepreneurs
- Be effective team contributors and/or leaders, including in multi-disciplinary environments, in professional fields related to computer science
- Serve their profession in a socially responsible, ethical, and innovative manner to meet the computing needs of industry, academia, and/or society
- Continue to develop and grow through lifelong learning, and successfully adapt to continuous advancements in computing technology

ABET Accreditation

The BS program in Computer Science is accredited by the Computing Accreditation Commission of ABET, www.abet.org.

Academic Progress in Computer Science

In order to make satisfactory academic progress as a Computer Science major, a student must meet all University requirements for academic progress, and must earn a grade of C or better in each of the "Major Requirements" courses. No "Major Requirements" course may be taken until the student has obtained a grade of C or better in each of the prerequisites. If a higher grade requirement is stated for a specific course, the higher requirement applies.

Bachelor of Science in Computer Science

[Degree Requirements](#) (124 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Farokh B. Bastani, Sergey Bereg, Ovidiu Daescu, Yvo G. Desmedt, Ding-Zhu Du, Paul Fishwick, Xiaohu Guo, Gopal Gupta, Kevin Hamlen, Sanda M. Harabagiu, Dung T. Huynh, Jason Jue, Latifur Khan, Andrian Marcus, Neeraj Mittal, Dan I. Moldovan, Sriraam Natarajan, Yu-Chung (Vincent) Ng, Tien Nguyen, Simeon C. Ntafos, Ravi Prakash, Kamil Sarac, Haim Schweitzer, Bhavani Thuraisingham, W. Eric Wong, Weili Wu, I-Ling Yen, Rym Zalila-Wenkstern

Associate Professors: Feng Chen, Lawrence Chung, Jorge A. Cobb, Vibhav Gogate, Benjamin Raichel, Nicholas Ruozzi

Assistant Professors: Xinya Du, Emily Kyle Fox, Yunhui Guo, Shuang Hao, Rishabh Iyer, Kangkook Jee, Chung Hwan Kim, Jin Kim, Jessica Ouyang, Yapeng Tian, Shiyi Wei, Yu Xiang, Wei Yang

Professors Emeriti: R. Chandrasekaran, Zygmunt Haas, Ivor P. Page, William J. Pervin, Balaji Raghavachari, Ivan Hal Sudborough, Subbarayan Venkatesan, Kang Zhang

Professors of Instruction: Ebru Cankaya, John Cole, Timothy (Tim) Farage, Shyam Karrah, Pushpa Kumar, Nhut Nguyen, Greg Ozbirn, Miguel Razo-Razo, Jeyakesavan (Jey) Veerasamy

Associate Professors of Instruction: Sridhar Alagar, Gordon Arnold, Anjum Chida, Wei Pang Chin, Bhadrachalam Chitturi, Chris I. Davis, Karen Doore, Neeraj Gupta, Khiem Le, Anarag Nagar, Mehra Nouroz Borazjany, Jalal Omer, Mark Paulk, Jason W. Smith, Laurie Thompson, James Willson, Nurcan Yuruk

Assistant Professors of Instruction: Eric Becker, Scott Dollinger, Serdar Erbatur, Ranran Feng, Omar Hamdy, Gity Karami, Kamran Khan, Karen Mazidi, Richard K. Min, Priya Narayanasami, Brian Ricks, Elmer Salazar, Meghana Satpute, Nidhiben Solanki, Srimathi Srinivasan, Yi Zhao

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

[RHET 1302](#) College Writing II¹

[ECS 2390](#) Professional and Technical Communication¹

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2413](#) Differential Calculus²

or [MATH 2417](#) Calculus I²

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics³

[PHYS 2326](#) Electromagnetism and Waves³

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

and a 3 semester credit hour [Government/Political Science Core](#) course

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2413](#) Differential Calculus²

or [MATH 2417](#) Calculus I²

[MATH 2414](#) Integral Calculus²

or [MATH 2419](#) Calculus II²

[PHYS 2125](#) Physics Laboratory I³

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 69 semester credit hours

Major Preparatory Courses: 24 semester credit hours beyond Core Curriculum

[ECS 1100](#) Introduction to Engineering and Computer Science⁴

[ECS 1210](#) Foundations in Engineering and Computer Science I

[CS 1436](#) Programming Fundamentals

[CS 1337](#) Computer Science I

[CS 2305](#) Discrete Mathematics for Computing I⁵

[CS 2336](#) Computer Science II

[CS 2340](#) Computer Architecture

[GOVT 2305](#) American National Government

[MATH 2418](#) Linear Algebra

[RHET 1302](#) College Writing II¹

[PHYS 2125](#) Physics Laboratory I³

[PHYS 2126](#) Physics Laboratory II

[PHYS 2325](#) Mechanics³

[PHYS 2326](#) Electromagnetism and Waves³

MATH Sequence - Students may choose one of the following sequences:

I. [MATH 2413](#) Differential Calculus²

and [MATH 2414](#) Integral Calculus²

or

II. [MATH 2417](#) Calculus I²

and [MATH 2419](#) Calculus II²

Major Core Courses: 36 semester credit hours beyond Core Curriculum

[ECS 2390](#) Professional and Technical Communication¹

[CS 3162](#) Professional Responsibility in Computer Science and Software Engineering

[CS 3341](#) Probability and Statistics in Computer Science and Software Engineering⁶

[CS 3345](#) Data Structures and Foundations of Algorithmic Analysis

[CS 3354](#) Software Engineering

[CS 3377](#) Systems Programming in UNIX and Other Environments

[CS 4141](#) Digital Systems Laboratory

[CS 4337](#) Programming Language Paradigms

[CS 4341](#) Digital Logic and Computer Design

[CS 4347](#) Database Systems

[CS 4348](#) Operating Systems Concepts

[CS 4349](#) Advanced Algorithm Design and Analysis

[CS 4384](#) Automata Theory

[CS 4485](#) Computer Science Project

Major Technical Electives: 9 semester credit hours

PCS technical electives are 4000 level courses in computer science or closely related areas approved by the student's academic advisor. The following courses may be used as technical electives without the explicit approval of an advisor:

[CS 4314](#) Intelligent Systems Analysis

[CS 4315](#) Intelligent Systems Design

[CS 4332](#) Foundations of Programming Video Games

[CS 4334](#) Numerical Analysis

[CS 4336](#) Advanced Java

[CS 4339](#) Web Programming Languages

[CS 4352](#) Foundations of Human-Computer Interaction

[CS 4361](#) Computer Graphics

[CS 4365](#) Artificial Intelligence

[CS 4375](#) Foundations of Machine Learning

[CS 4376](#) Object-Oriented Design

[CS 4386](#) Compiler Design

[CS 4389](#) Data and Applications Security

[CS 4390](#) Computer Networks

[CS 4391](#) Foundations of Computer Vision

[CS 4392](#) Computer Animation

[CS 4393](#) Computer and Network Security

[CS 4394](#) Implementation of Modern Operating Systems

[CS 4395](#) Human Language Technologies

[CS 4396](#) Networking Laboratory

[CS 4397](#) Embedded Computer Systems

[CS 4398](#) Digital Forensics

[CS 4399](#) Thesis Research in Computer Science

[CS 4459](#) Cyber Attack and Defense Laboratory

[EE 4325](#) Introduction to VLSI Design

[SE 4351](#) Requirements Engineering

[SE 4352](#) Software Architecture and Design

[SE 4367](#) Software Testing, Verification, Validation and Quality Assurance

[SE 4381](#) Software Project Planning and Management

Any other organized CS 4000-level course

If eligible, graduate courses may also be used to satisfy CS technical elective requirements.

III. Elective Requirements: 13 semester credit hours

Free Electives: 13 semester credit hours

Both lower- and upper-division courses may count as free electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Semester credit hours fulfill the communication component of the Core Curriculum.
2. Three semester credit hours of Calculus are counted under Mathematics Core, and five semester credit hours of Calculus are counted as Component Area Option Core.
3. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted as Component Area Option Core.
4. Transfer students with sufficient background may petition to substitute upper-division semester credit hours in the major for this class.
5. BS in Data Science students can substitute MATH 3315 for CS 2305.
6. BS in Data Science students can substitute STAT 3355 for CS 3341.

Fast Track Baccalaureate/Master's Degrees

In response to the need for post-baccalaureate education in the exciting field of computer science, a Fast Track program is available to well-qualified UT Dallas undergraduate students. Qualified seniors may take up to 15 graduate semester credit hours that may be used to complete the baccalaureate degree and also to satisfy requirements for the master's degree. Interested students should see the Associate Head for Graduate Studies in the Computer Science Department for specific requirements.

Honors Programs

The Department of Computer Science offers an Honors Program called Computer Science Computing Scholars (CS²). (CS²) is an intense Bachelor of Science in Computer Science Degree Program created for exceptionally gifted students who wish to pursue a demanding course of study enriched throughout with research experiences. The Computing Scholars Program has a specially designed curriculum. Courses integrate discussion of current research, recent discoveries, and open problems into a rich logical progression of firmly related topics. Course numbers for the

Core Curriculum Requirements and Major Requirements are the same as those for the Bachelor of Science in Computer Science, but Computing Scholars take honors versions of the following courses: [CS 1200](#), [CS 2305](#), [CS 2340](#), [CS 3341](#), [CS 3345](#), [CS 3354](#), [CS 4141](#), [CS 4337](#), [CS 4341](#), [CS 4348](#), [CS 4349](#), and [CS 4384](#).

Admission to the program is mainly by nomination and invitation. Those invited to join the Computing Scholars Honors Program will have successfully completed a full and challenging high school curriculum, will have achieved high scores on the SAT or ACT tests, and will be about to graduate from high school, or equivalent, with high class rank.

Successful participants will graduate with the added distinction of a Computing Scholars Honors Diploma.

For more information about this program students should contact the Computer Science Department leadership.

Minors

Students will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average).

Minor in Computer Science

21 semester credit hours

Students majoring in Computer Engineering, Computer Science or Software Engineering cannot add a minor in Computer Science.

A minor in Computer Science requires 21 semester credit hours earned through the following courses:

[CS 1337](#) Computer Science I

[CS 2305](#) Discrete Mathematics for Computing I

[CS 2336](#) Computer Science II

[CS 3345](#) Data Structures and Foundations of Algorithmic Analysis

[CS 3354](#) Software Engineering

Any 4000-level organized CS classes adding up to at least six semester credit hours

Minor in Cyber Security

30 semester credit hours

A minor in Cyber Security requires at least 30 semester credit hours earned through the following courses:

[CS 1337](#) Computer Science I

[CS 2305](#) Discrete Mathematics for Computing I

[CS 2336](#) Computer Science II

[CS 3345](#) Data Structures and Foundations of Algorithmic Analysis

[CS 4347](#) Database Systems

[CS 4348](#) Operating Systems Concepts

[CS 4390](#) Computer Networks

Any three courses out of the following set of courses:

[CS 4389](#) Data and Applications Security

[CS 4393](#) Computer and Network Security

[CS 4398](#) Digital Forensics

[CS 4459](#) Cyber Attack and Defense Laboratory

Certificates

Certificate in Cyber Defense

9 semester credit hours

A certificate in Cyber Defense is offered by the Department of Computer Science.

Required Courses:

Complete at least three courses (as well as any required prerequisites) from the following set of courses:

[CS 4389](#) Data and Applications Security

[CS 4393](#) Computer and Network Security

[CS 4398](#) Digital Forensics

[CS 4459](#) Cyber Attack and Defense Laboratory

Note: Students that complete the minor in Cyber Security will not be awarded a certificate in Cyber Defense.

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